

Lesson 3 · Quadratics and the Discriminant

The one idea: get = 0 first

A product is zero only when one of its factors is zero. So $(x - 3)(x + 2) = 0$ gives $x = 3$ or $x = -2$, but $(x - 3)(x + 2) = 6$ tells you nothing about either bracket. **Rearrange to $ax^2 + bx + c = 0$ before you do anything else.** Then three tools work: factorise, the formula, and the discriminant $b^2 - 4ac$, which counts the solutions before you find them.

1 Solving a quadratic

Write it in this order

1. Rearrange to $ax^2 + bx + c = 0$. Divide through by a common factor if there is one.
2. Try to factorise. If it factorises, set each bracket to zero.
3. If it does not, use the formula (on the formula sheet), with a, b, c in brackets:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

4. Give exact answers (surds) when asked; otherwise round at the end.

Example 1. Solve $x(x + 2) = 15$.

$$\begin{aligned}x^2 + 2x - 15 &= 0 \\(x + 5)(x - 3) &= 0 \\x &= -5 \text{ or } x = 3\end{aligned}$$

Not $x = 15$: the right side was not zero.

Example 2. Solve $2x^2 - 5x - 1 = 0$.

$a = 2, b = -5, c = -1$:

$$x = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(2)(-1)}}{2(2)} = \frac{5 \pm \sqrt{33}}{4}$$

$x = 2.686$ or $x = -0.186$

Example 3. Solve $x^2 - 6x + 2 = 0$ by completing the square.

$$\begin{aligned}x^2 - 6x &= -2 \\(x - 3)^2 - 9 &= -2 \\(x - 3)^2 &= 7 \\x &= 3 \pm \sqrt{7}\end{aligned}$$

(u)

(u)

Halve the x coefficient ($-6 \rightarrow -3$) and take away its square.

Example 4. Solve $x^4 - 13x^2 + 36 = 0$.

Let $u = x^2$: $u^2 - 13u + 36 = (u - 4)(u - 9) = 0$, so $x^2 = 4$ or $x^2 = 9$: $x = \pm 2$ or ± 3 .

(u)

(r)

2 From the roots back to the equation

A root $x = r$ comes from the factor $(x - r)$; a root $x = \frac{p}{q}$ comes from the factor $(qx - p)$, which keeps the coefficients whole.

Example 5. A quadratic has roots $\frac{3}{5}$ and -2 . Find it with whole-number coefficients.

$$\begin{aligned}x &= \frac{3}{5} \Rightarrow (5x - 3) \\x &= -2 \Rightarrow (x + 2) \\(5x - 3)(x + 2) &= 5x^2 + 7x - 6 = 0\end{aligned}$$

(r)

Example 6. A parabola crosses the x -axis at -1 and 5 and passes through $(0, -10)$.

$$\begin{aligned}y &= a(x + 1)(x - 5) \\-10 &= a(1)(-5) \Rightarrow a = 2 \\y &= 2x^2 - 8x - 10\end{aligned}$$

(r)

3 The discriminant

$$\Delta = b^2 - 4ac$$



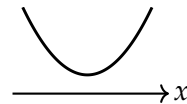
$$b^2 - 4ac > 0$$

two different real roots



$$b^2 - 4ac = 0$$

one repeated root (equal roots)



$$b^2 - 4ac < 0$$

no real roots

“Real roots” (without “different”) means $b^2 - 4ac \geq 0$.

Write it in this order (find k)

1. Read the condition and choose the sign: > 0 , $= 0$, < 0 or ≥ 0 .
2. Write a , b , c , each **in brackets**, with k wherever it appears.
3. Substitute into $b^2 - 4ac$ and solve the equation or inequality. Dividing an inequality by a negative number **turns it round**.

Example 7. $x^2 + kx + 16 = 0$ has equal roots. Find k .

$$k^2 - 4(1)(16) = 0 \Rightarrow k = \pm 8$$

(u)

Example 8. $2x^2 + 6x + p = 0$ has two different real roots. Find p .

$$36 - 8p > 0 \Rightarrow -8p > -36 \Rightarrow p < 4.5$$

(r)

Example 9. $kx^2 + 4x + (k - 3) = 0$ has equal roots. Find k .

$$4^2 - 4(k)(k - 3) = 0$$

$$k^2 - 3k - 4 = 0$$

$$k = 4 \text{ or } k = -1$$

(r)

4 Proving something about the roots

Example 10. Show that one solution of $x^2 - 5kx + 4k^2 = 0$ ($k \neq 0$) is four times the other.

$$(x - k)(x - 4k) = 0 \Rightarrow x = k \text{ or } x = 4k, \text{ and } 4k = 4 \times k.$$

(r)

Or by the formula: $\Delta = 25k^2 - 16k^2 = 9k^2$, so $x = \frac{5k \pm 3k}{2} = 4k \text{ or } k$. **Finish with the sentence the question asks for.**

Always real roots. To show an equation has real roots for **every** k , write Δ as a square: $(k + 2)^2 - 8k = (k - 2)^2 \geq 0$.

5 Versions the marker won't accept

Question	Answer given	Why it falls short
Roots $-\frac{2}{3}$ and 4; give $ax^2 + bx + c = 0$ with whole numbers	$x^2 - \frac{10}{3}x - \frac{8}{3} = 0$	Correct but not whole numbers: Achievement evidence only. Multiply by 3: $3x^2 - 10x - 8 = 0$.
$y = 0$ has exactly one solution	$144 - 8k > 0$	Wrong condition: one solution means $\Delta = 0$.
Show one solution is twice the other	$x = k \text{ or } x = 2k$	Add the conclusion: “so one solution is twice the other”.

6 Ways to lose marks

From the wording of the marking schedules and from teaching observation; numbered for marking.

- L17 Signs wrong when forming an equation from its roots.** A root of 4 gives $(x - 4)$, a root of $\frac{2}{3}$ gives $(3x - 2)$.
- L18 Coefficients not whole numbers when asked.** Clear fractions before expanding, or multiply through at the end.
- L19 Not set equal to zero before factorising.** $x(x + 3) = 10$ does not give $x = 10$.
- L20 Wrong discriminant condition.** Two different: > 0 . Equal or exactly one: $= 0$. None: < 0 . Real: ≥ 0 .
- L21 Negative b or k not bracketed.** $(-5)^2 = 25$, but $-5^2 = -25$. Brackets round b , and round k terms.

7 Quick review

Factorising (Lesson 1)

$$x^2 + 5x + 6 \quad (x + 2)(x + 3)$$

$$x^2 - 9 \quad (x - 3)(x + 3)$$

$$2x^2 + 5x + 2 \quad (2x + 1)(x + 2)$$

$$3x^2 - 6x \quad 3x(x - 2)$$

Linear inequalities

$$3x - 2 > 7 \quad x > 3$$

$$12 - 4k < 0 \quad -4k < -12, \text{ so } k > 3$$

$$-2p \geq 8 \quad p \leq -4$$

Multiply or divide by a negative: turn the sign round.